Camila Paz Guajardo Vásquez

Camila Guajardo Vásquez | cguajardo@dim.uchile.cl

PhD student
Department of Mathematical Engineering, Universidad de Chile.

My research interests are geometric topology, geometric analysis and hyperbolic geometry.

In my Master's thesis, I implemented a computational routine to obtain bounds on the systolic length and kissing number of compact Riemann surfaces, starting form an abstract group action.

Education

- PhD in Engineering Sciences, Mathematical Modeling — 2026 – present
  Institution: Faculty of Physical and Mathematical Sciences, Universidad de Chile
- Master of Science in Mathematics — 2024 – 2026
  Thesis: *On the action of the automorphism group of a compact Riemann surface on its set of systoles*
  Supervisor: PhD. Anita María Rojas Rodríguez
  Institution: Faculty of Sciences, Universidad de Chile
- Bachelor's Degree in Mathematics — 2020 – 2023
  Undergraduate thesis: *Mapping class groups of surfaces*
  Supervisor: PhD. Mauricio Bustamante Londoño
  Institution: Faculty of Mathematics, Pontificia Universidad Católica de Chile

Grants and Awards

- ANID National Doctorate Schorlarship (Beca de Doctorado Nacional) — 2026
- ANID National Master's Schorlarship (Beca de Magíster Nacional) — 2025
- Academic Excellence Award, Class of 2023 (Bachelor's in Mathematics) — awarded in 2024

Research Experience

- Thesis Student, Fondecyt Project No. 12340034 — 2024
  Topic: Group actions on compact Riemann surfaces and their effect on the set of closed geodesics.
  Supervisor: Prof. Anita Rojas
- Undergraduate thesis — 2023
  Topic: *Mapping Class Group* of surfaces
  Supervisor: Prof. Mauricio Bustamante Londoño
- Undergraduate Research Projects (Vicerrectoría de Investigación, PUC) — 2022–2023
  - Riemann-Hurwitz formula and applications, with Prof. Natalia García Fritz
  - More theory and applications in topology, with Prof. Mauricio Bustamante Londoño
  - Geometry of plane curves, with Prof. Giancarlo Urzúa

Academic Activities (Selected)

- XXII Escola de Geometria Diferencial. Teresina, Piauí, Brazil — 2026
  Poster presentation. Topic: An algorithm to provide systolic bounds from hyperbolic polygons.

- XXXVII Jornadas de Matemática de la Zona Sur, Universidad del Bío-Bío, Concepción — 2026
  Short communication. Topic: Systolic bounds from Hyperbolic Polygons.

- Organizer, Student Lectures Series in Riemannian Geometry, PUC — 2025 (2nd Term)
  Topics: Smooth manifolds, tangent spaces, Riemannian metrics, model spaces, connections, geodesics, and curvature.
  Faculty Advisor: Prof. Pedro Gaspar

- PhD's seminar, Universidad de Chile — 2026
  Topic: An Algorithm to Count Systoles in Compact Hyperbolic Surfaces.

- Student's seminar, Universidad de Talca — 2025
  Topic: Systolic bounds obtained from hyperbolic polygons

- Student's seminar, Universidad de Chile — 2024–2025
  Talks included *Concrete surfaces using SageMath* and An Introduction to Morse theory

- Number theory Seminar, PUC — 2022-2023
  Talks on additive fibers and elliptic surfaces over $\mathbb{P}^1$,and elliptic curves and their rational points

Selected Coursework

- POST664 – Analysis II (Prof. Nikola Kamburov) — 6.4/7.0
  Contents: Banach and Hilbert spaces, Baire and Hahn–Banach theorems, Lax–Milgram theorem, spectral theory, Sobolev spaces, distributions

- POST5900 – Introduction to Riemann surfaces and algebraic curves (Prof. Anita Rojas) — 7.0/7.0
  Contents: Riemann surfaces, algebraic curves in $\mathbb{P}^2(\mathbb{C})$, meromorphic functions, holomorphic maps, Riemann–Hurwitz formula

- CSDMATE157 – Complex tori (Prof. Robert Auffarth) — 6.1/7.0
  Contents: Torus homomorphisms, Néron–Severi group, dual tori, extensions, subtori complements

- CS07MM0019 – Geometry and topology of manifolds (Prof. Mauricio Bustamante) — 6.7/7.0
  Contents: Smooth manifolds, immersions and submersions, Sard's theorem, transversality

- MAT2555 – Functional analysis (Prof. Nikola Kamburov) — 6.3/7.0
  Contents: Banach spaces, Baire's and Hahn–Banach's theorems, Hilbert spaces, Fourier series, spectral theory of self-adjoint compact operators

- MAT2850 – Algebraic topology (Prof. Siddarth Mathur) — 6.3/7.0
  Contents: Covering spaces, fundamental groups, singular and cellular homology

- MAT2335 – Introduction to Algebraic geometry (Prof. Natalia García) — 6.0/7.0
  Contents: Algebraic curves, affine and projective varieties, birational morphisms

- MAT2305 – Differential Geometry, con (Prof. Mauricio Bustamante) — 6.4/7.0
  Contents: Curves in $\mathbb{R}^3$, fundamental forms, curvature, geodesics, Gauss–Bonnet's theorem

Teaching experience (Selected)

- Instructor. Department of Mathematics, Universidad Andrés Bello
  - FMMP001 – Mathematical thinking 2026
- Teaching Assistant. Faculty of Mathematics, Pontificia Universidad Católica de Chile
  - MPG3403 – Geometry and topology of manifolds, Prof. Pedro Gaspar 2025
  - MAT2704/MAT270I – Complex analysis, Prof. Martin Chuaqui 2024
  - MAT1389 – Geometry I, Profs. María José Moreno and María Fernanda Espinal 2023
  - MAT2218 – Abstract Algebra II, Prof. Ricardo Menares 2023
  - MAT1204 – Introduction to Algebra, Prof. Amal Taarabt 2022
- Teaching Assistant. Department of Mathematics, Universidad de Chile
  - CVER004 – Algebra and Geometry II, Prof. Anita Rojas 2025
  - MCLM110/MC110 – Algebra in Geometry I, Prof. Nelda Jaque Tamblay 2024

Outreach and service

- Coach, Chilean team for International Mathematical Olympiads 2021 - present
- Problem grader, Chilean National Mathematical Olympiad 2021 - present
- Invited speaker, Inspiring talks in NiñasPro 2026
- Mathematics Festival Monitor, Puerto de Ideas Antofagasta 2023
- Member, Academic Committee CMat (Campeonato de Matemáticas) 2020 - 2022
- Problem Coordinator, 1st Pan–American Girls' Mathematical Olympiad 2021

Mathematical Olympiads' Participation

- Ibero-American MO, Bronze Medal Mexico, 2019
- International MO, Honorable Mention UK, 2019
- European Girls' MO, Bronze Medal Ukraine, 2019
- International MO Romania, 2018
- South Cone MO, Honorable Mention Ecuador, 2017
- International MO Brazil, 2017